Modern Cryptography and AI/ML Security

LECTURE 32

Lattices

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1 From Vector Spaces to Lattices

A vector space over $\mathbb{R}$ allows arbitrary real-valued linear combinations of its basis vectors. A lattice, in contrast, is a *discrete* set of points obtained using integer-valued combinations of linearly independent vectors. This distinction is fundamental in lattice-based cryptography.

1.1 Linear span

Let $B = (\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n)$ be a collection of vectors in $\mathbb{R}^m$. Its real span is

$$\text{span}_{\mathbb{R}}(B) = \left\{ \sum_{i=1}^{n} x_i \boldsymbol{b}_i : x_i \in \mathbb{R} \right\}.$$

If the vectors are linearly independent, they form a basis for the subspace they span. When $n = m$ and the vectors are independent, they form a basis of $\mathbb{R}^n$.

Remark

The real span contains infinitely many points between any two distinct points. It is therefore a continuous object, not a lattice.

1.2 Lattice definition

Lattice

Let $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n \in \mathbb{R}^m$ be linearly independent vectors, where $n \leq m$. The lattice generated by these vectors is

$$\boxed{\mathcal{L}(B) = \left\{ \sum_{i=1}^{n} z_i \boldsymbol{b}_i : z_i \in \mathbb{Z} \right\},}$$

where $B = [\boldsymbol{b}_1 \ \boldsymbol{b}_2 \ \cdots \ \boldsymbol{b}_n]$ is the basis matrix. Equivalently,

$$\mathcal{L}(B) = \{B\boldsymbol{z} : \boldsymbol{z} \in \mathbb{Z}^n\}.$$

The coefficients must be integers. Thus, the lattice consists of discrete points rather than all points in the real span.

Remark

The rank of the lattice is n, the number of linearly independent basis vectors. Its ambient dimension is m. A full-rank lattice in $\mathbb{R}^m$ has rank m.

1.3 A two-dimensional example

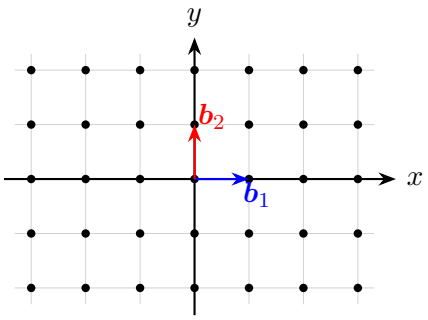
Consider

$$\boldsymbol{b}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \boldsymbol{b}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Then

$$\mathcal{L}(B) = \left\{ \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} : z_1, z_2 \in \mathbb{Z} \right\} = \mathbb{Z}^2.$$

This is the familiar square grid of integer-coordinate points.



Every lattice point has the form

$$\boldsymbol{x} = z_1\boldsymbol{b}_1 + z_2\boldsymbol{b}_2, \qquad z_1, z_2 \in \mathbb{Z}.$$

For example, $(2, -1)^T = 2\boldsymbol{b}_1 - \boldsymbol{b}_2$ is a lattice point, whereas $(1/2, 1/3)^T$ is not.

2 Basis, Independence, and Coordinates

2.1 Why must the basis vectors be independent?

The basis vectors of a rank-n lattice must be linearly independent. If they are dependent, the representation of a point by integer coefficients may fail to be unique, and the set may not have the intended rank.

Example

Let

$$\boldsymbol{b}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \boldsymbol{b}_2 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$$

These vectors are linearly dependent. The point $(2, 0)^T$ has two representations:

$$2\boldsymbol{b}_1 = \boldsymbol{b}_2.$$

The generated set is just the one-dimensional integer lattice on the horizontal axis, not a two-dimensional lattice.

Theorem 2.1 (Uniqueness of coordinates). *If the columns of $B \in \mathbb{R}^{m\times n}$ are linearly independent, then for every $\boldsymbol{x} \in \operatorname{span}_{\mathbb{R}}(B)$ there is a unique vector $\boldsymbol{c} \in \mathbb{R}^n$ such that $\boldsymbol{x} = B\boldsymbol{c}$. In particular, every lattice point has a unique integer-coordinate representation relative to a fixed basis.*

Proof. Suppose $B\boldsymbol{c} = B\boldsymbol{d}$. Then $B(\boldsymbol{c} - \boldsymbol{d}) = \boldsymbol{0}$. Since the columns of B are linearly independent, the nullspace of B is trivial. Hence $\boldsymbol{c} - \boldsymbol{d} = \boldsymbol{0}$, so $\boldsymbol{c} = \boldsymbol{d}$. □

2.2 A lattice is infinite but discrete

For any nonzero basis vector $\boldsymbol{b}_1$, all integer multiples

$$\ldots, -2\boldsymbol{b}_1, -\boldsymbol{b}_1, \boldsymbol{0}, \boldsymbol{b}_1, 2\boldsymbol{b}_1, \ldots$$

belong to the lattice. Thus, a lattice contains infinitely many points. However, these points do not fill the real span: there are gaps between nearby lattice points.

3 Different Bases for the Same Lattice

A lattice has infinitely many possible bases. The basis is not unique, even though the lattice itself is unchanged.

Example

Consider

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad B' = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

The columns of B' are

$$\boldsymbol{b}'_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \boldsymbol{b}'_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Both matrices generate $\mathbb{Z}^2$, since

$$\begin{pmatrix} x \\ y \end{pmatrix} = (x - 2y)\boldsymbol{b}'_1 + y\boldsymbol{b}'_2$$

for every $x, y \in \mathbb{Z}$. Therefore $\mathcal{L}(B) = \mathcal{L}(B') = \mathbb{Z}^2$.

3.1 Unimodular matrices

Unimodular matrix

A square integer matrix $U \in \mathbb{Z}^{n\times n}$ is called unimodular if

$$\det(U) = +1 \quad \text{or} \quad \det(U) = -1.$$

Equivalently, U^{-1} is also an integer matrix.

Theorem 3.1 (Change of basis). *Let $B \in \mathbb{R}^{m\times n}$ have linearly independent columns and let $U \in \mathbb{Z}^{n\times n}$ be unimodular. Then*

$$\boxed{\mathcal{L}(BU) = \mathcal{L}(B).}$$

Conversely, any two bases of the same rank-n lattice are related by a unimodular integer matrix: if B and B' are basis matrices for the same lattice, then $B' = BU$ for some unimodular U.

Proof. Since U is an integer matrix, $U\boldsymbol{z} \in \mathbb{Z}^n$ for every $\boldsymbol{z} \in \mathbb{Z}^n$. Therefore

$$\mathcal{L}(BU) = \{BU\boldsymbol{z} : \boldsymbol{z} \in \mathbb{Z}^n\} \subseteq \{B\boldsymbol{y} : \boldsymbol{y} \in \mathbb{Z}^n\} = \mathcal{L}(B).$$

Because U is unimodular, U^{-1} is also an integer matrix. Reversing the argument gives $\mathcal{L}(B) \subseteq \mathcal{L}(BU)$. Thus equality holds. □

Remark

A change of basis changes the coordinate description and the geometry of the displayed basis vectors, but it does not change the lattice points.

4 The Shortest Vector Problem

A central computational question is: given a lattice basis, can we find the shortest nonzero lattice vector?

First minimum

For a lattice $\mathcal{L} \subset \mathbb{R}^m$, its first successive minimum is

$$\lambda_1(\mathcal{L}) = \min_{\boldsymbol{v}\in\mathcal{L}\setminus\{\boldsymbol{0}\}} \|\boldsymbol{v}\|_2 .$$

It is the Euclidean length of a shortest nonzero lattice vector.

Shortest Vector Problem (SVP)

Given a basis B of a lattice $\mathcal{L}(B)$, find a nonzero vector

$$\boldsymbol{v} \in \mathcal{L}(B) \quad \text{such that} \quad \|\boldsymbol{v}\|_2 = \lambda_1(\mathcal{L}(B)).$$

This is the exact Shortest Vector Problem.

4.1 An example

For the standard lattice $\mathbb{Z}^2$, the shortest nonzero vectors are

$$(1,0), \quad (-1,0), \quad (0,1), \quad (0,-1),$$

and hence $\lambda_1(\mathbb{Z}^2) = 1$.

For a general basis, the shortest vector need not be one of the basis vectors. For example, a basis may contain long, nearly parallel vectors whose difference is much shorter.

4.2 Approximate SVP

Finding an exact shortest vector can be computationally difficult in high dimensions. An approximation version asks for a vector that is not necessarily shortest, but whose length is within a specified factor of the optimum.

Approximate SVP

For a factor $\gamma \geq 1$, the γ-approximate SVP asks for a nonzero vector $\boldsymbol{v} \in \mathcal{L}$ satisfying

$$\|\boldsymbol{v}\|_2 \leq \gamma\, \lambda_1(\mathcal{L}).$$

The value of γ specifies how much longer than the shortest vector the answer is allowed to be.

4.3 Algorithms and computational complexity

Lattice reduction algorithms seek short and relatively independent basis vectors.

- **LLL algorithm:** A polynomial-time lattice basis reduction algorithm. It produces a nonzero vector whose length is bounded by an exponential-in-dimension approximation factor (commonly stated as $2^{(n-1)/2}$ for a standard LLL guarantee).
- **BKZ algorithm:** A stronger blockwise lattice reduction method. Increasing the block size generally improves the quality of the reduced basis, but increases computational cost.
- **Exact algorithms:** Algorithms for exact SVP can have exponential running time in the dimension in known general approaches.

Remark

Polynomial-time approximation algorithms do not imply that exact SVP is easy. The approximation factor and the dimension are essential parts of the complexity statement.

5 The Dual Lattice

The dual lattice captures vectors whose inner products with all lattice vectors are integers.

Dual lattice

Let $\mathcal{L} \subset \mathbb{R}^m$ be a full-rank lattice. Its dual lattice is

$$\boxed{\mathcal{L}^* = \{\boldsymbol{y} \in \mathbb{R}^m : \langle \boldsymbol{y}, \boldsymbol{x} \rangle \in \mathbb{Z} \text{ for every } \boldsymbol{x} \in \mathcal{L}\}.}$$

If $B \in \mathbb{R}^{m \times m}$ is a basis matrix of a full-rank lattice, then

$$\boxed{\mathcal{L}(B)^* = B^{-T}\mathbb{Z}^m,} \qquad B^{-T} := (B^{-1})^T.$$

Thus, a basis matrix for the dual lattice is

$$B^* = (B^{-1})^T.$$

Proof. A vector $\boldsymbol{y}$ belongs to $\mathcal{L}(B)^*$ exactly when $\langle \boldsymbol{y}, B\boldsymbol{z} \rangle \in \mathbb{Z}$ for every $\boldsymbol{z} \in \mathbb{Z}^m$. This is equivalent to $B^T\boldsymbol{y} \in \mathbb{Z}^m$. Hence $\boldsymbol{y} = B^{-T}\boldsymbol{z}$ for some $\boldsymbol{z} \in \mathbb{Z}^m$. □

Example

For $\mathcal{L} = \mathbb{Z}^2$, the basis is $B = I_2$, so

$$\mathcal{L}^* = (I_2^{-1})^T\mathbb{Z}^2 = \mathbb{Z}^2.$$

The standard integer lattice is self-dual.

6 Determinant and the Volume of a Lattice

For a full-rank lattice $\mathcal{L}(B)$ with square basis matrix $B \in \mathbb{R}^{n \times n}$, the fundamental parallelepiped is

$$\mathcal{P}(B) = \{B\boldsymbol{t} : \boldsymbol{t} \in [0,1)^n\}.$$

Its volume is the absolute determinant of the basis matrix.

Determinant of a lattice

For a full-rank lattice $\mathcal{L}(B) \subset \mathbb{R}^n$, define

$$\boxed{\det(\mathcal{L}(B)) = |\det(B)|.}$$

This is also called the covolume of the lattice.

Theorem 6.1 (Basis independence of the determinant). *If $B' = BU$ for a unimodular matrix U, then*

$$|\det(B')| = |\det(B)|.$$

Consequently, the lattice determinant does not depend on the choice of basis.

Proof. Using multiplicativity of the determinant,

$$|\det(B')| = |\det(BU)| = |\det(B)|\,|\det(U)| = |\det(B)|.$$

□

For a full-rank lattice, the dual determinant satisfies

$$\boxed{\det(\mathcal{L}^*) = \frac{1}{\det(\mathcal{L})}.}$$

This follows from $\det(B^{-T}) = 1/\det(B)$.

Remark

For a rank-n lattice embedded in $\mathbb{R}^m$ with $m > n$, the corresponding volume is the n-dimensional volume of its fundamental parallelepiped, equal to $\sqrt{\det(B^T B)}$. The square-matrix determinant formula applies directly to full-rank lattices.

7 q-ary Lattices

Lattice-based cryptography frequently uses lattices defined through modular linear algebra. Let $q \geq 2$ be an integer, usually chosen prime in basic presentations, and let

$$A \in \mathbb{Z}_q^{n \times m}.$$

Here, $\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$ and all matrix-vector products involving A are computed modulo q.

The q-ary kernel lattice

The q-ary lattice associated with the modular matrix A is

$$\boxed{\Lambda_q^{\perp}(A) = \{\boldsymbol{x} \in \mathbb{Z}^m : A\boldsymbol{x} = \mathbf{0} \pmod q\}.}$$

It consists of all integer vectors mapped to zero by A modulo q.

The q-ary image lattice

The companion q-ary lattice is

$$\boxed{\Lambda_q(A) = \{\boldsymbol{A}^T\boldsymbol{s} + q\boldsymbol{z} : \boldsymbol{s} \in \mathbb{Z}^n,\ \boldsymbol{z} \in \mathbb{Z}^m\}.}$$

Equivalently, $\boldsymbol{s}$ may be taken modulo q, since changing $\boldsymbol{s}$ by a multiple of q changes $\boldsymbol{A}^T\boldsymbol{s}$ by a vector in $q\mathbb{Z}^m$.

Theorem 7.1 (Duality of q-ary lattices)**.** *With the definitions above,*

$$\boxed{\Lambda_q(A)^* = \frac{1}{q}\Lambda_q^{\perp}(A).}$$

Equivalently,

$$\Lambda_q^{\perp}(A)^* = \frac{1}{q}\Lambda_q(A),$$

under the usual full-rank assumptions.

Remark

The transpose in $A^T\boldsymbol{s}$ is necessary for dimensions to match: A is $n \times m$, so $A^T\boldsymbol{s}$ is an m-dimensional vector.

7.1 A small modular example

Let $q = 5$ and take

$$A = \begin{pmatrix} 1 & 2 \end{pmatrix} \in \mathbb{Z}_5^{1\times 2}.$$

Then

$$\Lambda_5^{\perp}(A) = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{Z}^2 : x_1 + 2x_2 = 0 \pmod 5 \right\}.$$

For example, $(1, 2)^T$ is in the kernel lattice because $1 + 2(2) = 5 = 0 \pmod 5$.

8 The Short Integer Solution Problem

The Short Integer Solution (SIS) problem is a central computational problem underlying many lattice-based cryptographic constructions, including hash functions and digital signatures.

SIS problem

Let $q \geq 2$, let $A \in \mathbb{Z}_q^{n \times m}$ be a given matrix, and let $\beta > 0$ be a norm bound. The SIS problem asks to find a nonzero vector

$$\boldsymbol{x} \in \mathbb{Z}^m$$

such that

$$\boxed{A\boldsymbol{x} = \mathbf{0} \pmod{q}, \qquad 0 < \|\boldsymbol{x}\|_2 \leq \beta.}$$

In other words, SIS asks for a short, nonzero vector in the lattice $\Lambda_q^{\perp}(A)$.

8.1 Why is SIS a lattice problem?

By definition,

$$\Lambda_q^{\perp}(A) = \{\boldsymbol{x} \in \mathbb{Z}^m : A\boldsymbol{x} = \mathbf{0} \pmod{q}\}.$$

The SIS solution must satisfy the same modular equation, together with a shortness condition. Therefore, SIS is precisely a search for a short nonzero vector in a particular q-ary lattice.

8.2 A simple example

Again take

$$q = 5, \qquad A = \begin{pmatrix} 1 & 2 \end{pmatrix}.$$

The vector

$$\boldsymbol{x} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

satisfies

$$A\boldsymbol{x} = 1 + 2(2) = 5 = 0 \pmod{5}.$$

Its Euclidean norm is

$$\|\boldsymbol{x}\|_2 = \sqrt{1^2 + 2^2} = \sqrt{5}.$$

Thus, for any bound $\beta \geq \sqrt{5}$, this vector is a valid SIS solution.

Remark

The example illustrates correctness, not hardness. Cryptographic parameters are chosen so that finding a sufficiently short nonzero solution is computationally infeasible for an adversary, under an appropriate hardness assumption.

9 Gaussian Sampling and Trapdoors: A Preview

Gaussian distributions play an important role in lattice cryptography. A discrete Gaussian distribution over a lattice assigns probabilities to lattice points according to their distance from a center. Informally, shorter vectors near the center receive higher probability.

For a lattice $\mathcal{L}$, center $\boldsymbol{c}$, and parameter $s > 0$, the discrete Gaussian distribution has the form

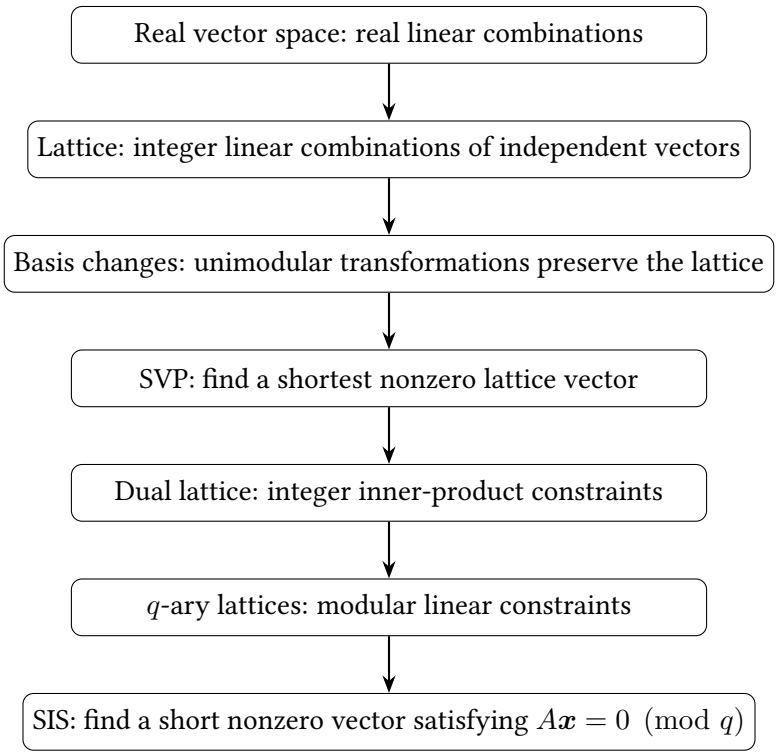
$$D_{\mathcal{L},s,\boldsymbol{c}}(\boldsymbol{x}) \propto \exp\left(-\frac{\pi \|\boldsymbol{x} - \boldsymbol{c}\|_2^2}{s^2}\right), \qquad \boldsymbol{x} \in \mathcal{L}.$$

The proportionality constant normalizes the probabilities to sum to one.

A *trapdoor* is secret auxiliary information that makes a computational task easy for its holder, while the corresponding public information does not reveal an efficient solution method to an attacker. In lattice cryptography, trapdoors are often used to sample short vectors satisfying specified modular linear constraints.

These ideas appear in lattice-based signatures, identity-based encryption, and other cryptographic constructions.

10 Summary and Conceptual Connections



Concept	**Main idea**
Lattice $\mathcal{L}(B)$	Integer combinations of linearly independent basis vectors.
Basis	A set of independent vectors that generates the entire lattice.
Unimodular matrix	An integer matrix with determinant ± 1; it changes the basis but not the lattice.
SVP	Find a shortest nonzero vector in a lattice.
Approximate SVP	Find a nonzero vector within a specified factor of the shortest length.
Dual lattice	Vectors having integer inner products with every lattice vector.
Determinant	Volume of a fundamental parallelepiped.
q-ary lattice	A lattice defined using modular linear equations.
SIS	Find a short, nonzero integer vector in the modular kernel of a given matrix.

11 Questions for Discussion

1. Why do we use integer coefficients rather than real coefficients in the definition of a lattice?
2. Can two different basis matrices generate exactly the same lattice? Explain using a unimodular matrix.
3. Why does linear independence guarantee unique coordinates relative to a fixed basis?
4. Why might the shortest lattice vector not be one of the supplied basis vectors?
5. What is the difference between exact SVP and approximate SVP?
6. How is the dual lattice defined, and what is its basis matrix?
7. Why is the determinant of a lattice independent of the chosen basis?
8. How does the modular equation $A\boldsymbol{x} = \mathbf{0} \pmod{q}$ define a lattice?
9. Explain why SIS can be viewed as a shortest-vector search in a q-ary lattice.
10. Why are lattice reduction algorithms relevant to the security analysis of lattice-based cryptography?

11.1 Definition of a Lattice

Let $\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n \in \mathbb{R}^m$ be linearly independent vectors. The lattice generated by these vectors is defined as

$$\mathcal{L}(B) = \left\{ \sum_{i=1}^{n} x_i \mathbf{b}_i \;\middle|\; x_i \in \mathbb{Z} \right\},$$

where $B = [\mathbf{b}_1 \mid \mathbf{b}_2 \mid \cdots \mid \mathbf{b}_n] \in \mathbb{R}^{m \times n}$ is called the basis matrix of the lattice.

Equivalently,

$$\boxed{\mathcal{L}(B) = \{B\mathbf{x} : \mathbf{x} \in \mathbb{Z}^n\}.}$$

A lattice is a discrete additive subgroup of $\mathbb{R}^m$ generated by integer linear combinations of linearly independent basis vectors.

Equivalent Bases. A lattice can have multiple bases. In particular, if $U \in \mathbb{Z}^{n \times n}$ is a unimodular matrix, i.e., $\det(U) = \pm 1$, then

$$\boxed{\mathcal{L}(B) = \mathcal{L}(BU).}$$

Thus, unimodular transformations change the basis representation without changing the underlying lattice.

11.2 Lattice Hardness Assumptions

The security of lattice-based cryptographic schemes relies on the computational hardness of certain problems associated with lattices.

1. Shortest Vector Problem (SVP). Given a lattice $\mathcal{L}(B)$, find a non-zero lattice vector $\mathbf{v} \in \mathcal{L}(B)$ having minimum Euclidean norm:

$$\boxed{\mathbf{v}^* = \arg \min_{\substack{\mathbf{v} \in \mathcal{L}(B) \\ \mathbf{v} \neq \mathbf{0}}} \|\mathbf{v}\|_2.}$$

2. Closest Vector Problem (CVP). Given a lattice $\mathcal{L}(B)$ and a target vector $\mathbf{t} \in \mathbb{R}^m$, find the lattice vector closest to $\mathbf{t}$:

$$\boxed{\mathbf{v}^* = \arg \min_{\mathbf{v} \in \mathcal{L}(B)} \|\mathbf{t} - \mathbf{v}\|_2.}$$

3. Learning With Errors (LWE). Let q be a modulus, n be a positive integer, and χ be an error distribution over $\mathbb{Z}$.

The LWE problem considers samples of the form

$$(\mathbf{a}, b), \qquad b = \langle \mathbf{a}, \mathbf{s} \rangle + e \pmod{q},$$

where

$$\mathbf{a} \leftarrow \mathbb{Z}_q^n, \quad \mathbf{s} \leftarrow \mathbb{Z}_q^n, \quad e \leftarrow \chi.$$

The objective is to distinguish LWE samples from uniformly random samples, or to recover the secret $\mathbf{s}$ from sufficiently many LWE samples.

The Learning With Errors problem is believed to be computationally hard for appropriately chosen parameters, even against quantum adversaries.

11.3 LWE-Based Public-Key Encryption

We describe a public-key encryption scheme based on the Learning With Errors (LWE) assumption.

The scheme consists of three algorithms:

$$\mathsf{PKE} = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec}).$$

Let n be the security parameter, q be a prime modulus, m be the number of samples, and χ be the error distribution.

1. Key Generation (KeyGen).

1. Choose a uniformly random matrix
$$A \leftarrow \mathbb{Z}_q^{m \times n}.$$
2. Choose a secret vector
$$\mathbf{s} \leftarrow \mathbb{Z}_q^n.$$
3. Sample an error vector
$$\mathbf{e} \leftarrow \chi^m.$$
4. Compute
$$\mathbf{b} = A\mathbf{s} + \mathbf{e} \pmod{q}.$$
5. Output the public and secret keys:
$$PK = (A, \mathbf{b}), \qquad SK = \mathbf{s}.$$

2. Encryption (Enc). Given a public key $PK = (A, \mathbf{b})$ and a message bit $\mu \in \{0, 1\}$:

1. Choose a random binary vector
$$\mathbf{r} \leftarrow \{0,1\}^m.$$
2. Compute
$$\mathbf{u} = A^T\mathbf{r} \pmod{q}.$$
3. Compute
$$v = \mathbf{b}^T\mathbf{r} + \mu \left\lfloor \frac{q}{2} \right\rfloor \pmod{q}.$$
4. Output the ciphertext:
$$CT = (\mathbf{u}, v).$$

3. Decryption (Dec). Given the secret key $SK = \mathbf{s}$ and ciphertext $CT = (\mathbf{u}, v)$:

1. Compute
$$z = v - \mathbf{s}^T\mathbf{u} \pmod{q}.$$
2. Recover the message bit using
$$\mu = \begin{cases} 0, & \text{if } z \text{ is closer to } 0, \\ 1, & \text{if } z \text{ is closer to } q/2. \end{cases}$$

Correctness. Since $\mathbf{b} = A\mathbf{s} + \mathbf{e}$, we obtain

$$\begin{aligned} z &= v - \mathbf{s}^T\mathbf{u} \\ &= \mathbf{b}^T\mathbf{r} + \mu \left\lfloor \frac{q}{2} \right\rfloor - \mathbf{s}^T A^T\mathbf{r} \\ &= \mathbf{e}^T\mathbf{r} + \mu \left\lfloor \frac{q}{2} \right\rfloor \pmod{q}. \end{aligned}$$

If the error satisfies

$$\left|\mathbf{e}^T\mathbf{r}\right| < \frac{q}{4},$$

using its centered representative modulo q, the receiver can correctly recover μ by nearest-center decoding.

Security Intuition. The public key contains noisy linear equations of the form

$$\mathbf{b} = A\mathbf{s} + \mathbf{e} \pmod{q}.$$

Although the public matrix A and vector $\mathbf{b}$ are known, recovering the secret $\mathbf{s}$ is computationally hard under the LWE assumption for suitable parameters.

The noise term $\mathbf{e}$ is essential: it prevents an adversary from directly solving the linear system over $\mathbb{Z}_q$.

12 GGH: A Lattice-Based Public Key Encryption Scheme

The Goldreich–Goldwasser–Halevi (GGH) cryptosystem, introduced in 1997, is an early public-key encryption scheme based on the hardness of lattice problems. Its security intuition relies on the difficulty of finding a short vector or solving the Closest Vector Problem (CVP) when the lattice is represented using a poor basis.

12.1 Hermite Normal Form (HNF)

Before describing the GGH cryptosystem, we introduce the Hermite Normal Form (HNF), which provides a canonical representation of an integer lattice.

Hermite Normal Form

A matrix $H \in \mathbb{Z}^{n\times n}$ is in lower triangular Hermite Normal Form if:

1. H is lower triangular:
$$h_{ij} = 0, \quad i < j.$$
2. All diagonal entries are positive:
$$h_{ii} > 0.$$
3. Entries below the diagonal satisfy
$$0 \le h_{ij} < h_{jj}, \quad i > j.$$

12.1.1 How to Compute HNF

The HNF of an integer matrix can be computed using elementary integer row operations:

1. Swap two rows.
2. Multiply a row by -1.
3. Add an integer multiple of one row to another row.

These operations correspond to multiplication by unimodular matrices and therefore preserve the lattice.

The objective is to transform the matrix into an upper triangular form and subsequently reduce the entries above the diagonal into their required ranges.

For example, consider

$$B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Its row HNF is

$$H = \begin{bmatrix} 3 & 0 \\ 1 & 1 \end{bmatrix}.$$

Indeed, applying the unimodular matrix

$$U = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

gives

$$H = UB = \begin{bmatrix} 3 & 3 \\ 2 & 1 \end{bmatrix},$$

which is not yet in HNF. Further integer row reduction gives the canonical HNF:

$$H = \begin{bmatrix} 3 & 0 \\ 1 & 1 \end{bmatrix}.$$

12.1.2 Why Is HNF Considered a Poor Basis in GGH?

A lattice can have several different bases, even though all these bases generate exactly the same lattice.

A good basis consists of relatively short and nearly orthogonal vectors. Such a basis makes the closest vector problem easier.

A poor basis, on the other hand, may contain long and highly correlated vectors, making it difficult to identify the closest lattice point.

In GGH, the public basis is chosen to be a poor basis, while the private basis is a good basis.

HNF can be used as a canonical public representation of the lattice. However, HNF is not inherently a poor basis. Its geometric quality depends on the lattice and its representation. In the GGH construction, the public basis is selected or generated so that it is difficult to use for solving CVP.

12.2 GGH Public Key Encryption

Let $B \in \mathbb{Z}^{n\times n}$ be a good basis of a lattice $\mathcal{L}$, and let $U \in \mathbb{Z}^{n\times n}$ be a unimodular matrix with large entries.

The public basis is constructed as

$$B_{\text{pub}} = UB.$$

Since U is unimodular,

$$\mathcal{L}(B_{\text{pub}}) = \mathcal{L}(B).$$

However, B_{pub} can be a poor basis even though B is a good basis.

The GGH cryptosystem consists of the following algorithms.

12.2.1 1. Key Generation

1. Generate a good lattice basis
$$B \in \mathbb{Z}^{n\times n}.$$
2. Generate a unimodular matrix
$$U \in \mathbb{Z}^{n\times n}, \qquad |\det(U)| = 1,$$
such that the resulting public basis is sufficiently poor.
3. Compute
$$B_{\text{pub}} = UB.$$

The keys are:

$$\boxed{\text{PK} = B_{\text{pub}}}$$

$$\boxed{\mathrm{SK} = B}$$

12.2.2 2. Encryption

Let $m \in \{0, 1\}$ be the plaintext bit.

The sender selects a random lattice point

$$v \in \mathcal{L}(B_{\mathrm{pub}})$$

and a small error vector

$$e \in \mathbb{Z}^n.$$

The ciphertext is computed as

$$\boxed{c = v + e + m \cdot t}$$

where $t \in \mathbb{Z}^n$ is a fixed public vector used to encode the message.

The error vector is selected such that its norm is sufficiently small relative to the decoding capability of the private basis.

12.2.3 3. Decryption

The receiver uses the private good basis B to find the closest lattice point to the ciphertext.

Let

$$v' = \mathrm{CVP}_B(c).$$

The receiver computes the residual

$$r = c - v'.$$

Since v is a lattice point, the receiver obtains

$$r = e + m \cdot t$$

provided that the decoding operation correctly recovers v.

The message is recovered by distinguishing the two possible residual classes corresponding to $m = 0$ and $m = 1$.

Thus,

$$\boxed{\mathrm{Dec}(c) = \begin{cases} 0, & r \text{ belongs to the class for } m = 0, \\ 1, & r \text{ belongs to the class for } m = 1. \end{cases}}$$

12.3 Security Intuition

The security of GGH relies on the difference between having a good lattice basis and having a poor lattice basis.

- The private key contains a good basis, which allows efficient approximate CVP decoding.
- The public key contains a poor basis, which makes decoding the closest lattice point computationally difficult.
- An adversary knows the public basis and ciphertext but does not know the private good basis.
- Recovering the closest lattice point using only the public basis is believed to be computationally difficult for appropriately chosen parameters.

The central security intuition can be summarized as follows:

$$\boxed{\text{Good Basis} \Longrightarrow \text{Efficient Decoding}}$$

$$\boxed{\text{Poor Basis} \Longrightarrow \text{Difficult Decoding}}$$

Therefore, the adversary faces the problem of decoding a lattice point using a poor basis, whereas the legitimate receiver can use the private good basis.

Important Security Remark. The original GGH construction is a historical lattice-based cryptosystem and should not be considered secure by modern standards. In particular, its security depends on more than the apparent poor geometric quality of the public basis. Structural attacks and lattice reduction techniques can recover useful short vectors or exploit the relationship between the public and private bases. Modern lattice-based cryptography instead relies heavily on constructions based on LWE, Ring-LWE, and Module-LWE. “‘